

Purnendu Tiwari

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Experience

Software Development Intern

FOSSEE, IIT Bombay

May 2026 – July 2026

Mumbai, India (Remote)

- Collaborated with a mentor and a 3-member team to modernize the Linux packaging pipeline for the eSim application across **AppImage** and **Flatpak** formats
- Fixed PyQt5 dependency crashes by bundling a self-contained Python runtime into the AppImage, removing all host-side setup requirements
- Resolved Qt platform **plugin failures** by building automatic X11/Wayland detection logic, fixing rendering across Linux desktop environments
- Rebuilt the Flatpak offline install pipeline by fixing broken package sources and pre-validating dependency wheels, enabling zero-network builds

Projects

sdchouse – Real-Time Virtual Space Platform

Live | [GitHub](#)

- Built a real-time multiplayer web app using **Next.js** allowing multiple users to move and interact together in a shared virtual space, with instant synchronization across all connected clients
- Built a **Socket.IO**-based signaling server on **Node.js/Express** to manage live game state (player positions, chat) and broker peer-to-peer connections between users
- Implemented **WebRTC** for direct audio/video calls between users without routing media through a server, keeping the system fast and easily scalable

RCE Engine – Remote Code Execution Platform

[GitHub](#)

- Built a **Remote Code Execution engine** that safely runs untrusted user code inside isolated **Docker** containers with resource limits, automatic timeout, and cleanup
- Designed an asynchronous job queue using **BullMQ** and **Redis** to process multiple code submissions at once, allowing the system to scale by adding more workers as load increases
- Built backend APIs with **Express** and **Prisma/PostgreSQL** to reliably track submission status and manage the full lifecycle of each code execution, from request to result
- Set up an automated CI/CD pipeline with **GitHub Actions** to run type-checks and integration tests on every change, catching bugs before deployment

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, Bash

Web Development: React JS, Node JS, Express JS, REST APIs, MongoDB, MySQL, PostgreSQL, Next.js, Prisma

Cloud & DevOps: AWS (S3, CloudFront, IAM, VPC, EC2, CloudFormation), Docker, Docker Compose, GitHub Actions, CI/CD, Nginx, Cloudflare, Vercel, Linux, Prometheus, grafana

Developer Tools: Git, GitHub, Bitbucket, Redis, BullMQ, Postman

Education

Vellore Institute of Technology (VIT)

B.Tech in Computer Science; CGPA: 8.99/10.0

Bhopal, India

2023 – 2027

Kendriya Vidyalaya No.1, Chhindwara

Higher Secondary (Class XII), CBSE – 92.4%

Chhindwara, India

2022

Certifications

AWS Certified Solutions Architect – Associate (SAA-C03) – Amazon Web Services

The Bits and Byte of Computer Networking – Google(Coursera)

Achievements & Co-curricular

Top Coder: Cleared inter-campus qualifier, securing selection among the top 363 out of 10,000+ students

Leadership: President, Software Development Club; led a 6-member team to host 5+ tech events for 300+ students

LeetCode Knight: Achieved Highest Contest Rating **1852** and Among the Top 5% of coders on LeetCode

GirlScript Summer of Code: Ranked **419th** among global open-source contributors

Music: Diploma in Classical Music (Raja Mansingh Tomar Music Arts University)